



Software Quality Assurance

Software Engineering

Software Quality Assurance

- ❖ **Software Quality Assurance (SQA)** refers to a systematic and planned set of actions aimed at ensuring software products and processes meet their specified requirements and standards. These include functionality, performance, design, and reliability standards.
- ❖ It's not just about testing the software after it's built but involves ensuring quality throughout the development lifecycle. This includes defined processes, standards, and methodologies as well as ongoing verification that the software is working as expected.



Software Quality Assurance Plan (SQAP)

- ❖ **Software Quality Assurance Plan (SQAP)** is a document used to guide the quality assurance activities during the software life cycle. It covers the following elements:
 - Purpose and Scope of the SQA Plan
 - Contents of an SQA Plan

- ❖ **Purpose and Scope of the SQA Plan:**
 - The **purpose** of an SQAP is to define the procedures, techniques, and tools to be used to assess and assure the quality of the software being developed or maintained.
 - The **scope** includes the software products, services, and activities that will be assessed, and the various phases of the software lifecycle that will be covered by SQA activities.

Software Quality Assurance Plan - Contents

- ❖ **Introduction:** Outlines the purpose, scope, and structure of the SQA plan.
- ❖ **Reference Documents:** Lists relevant standards, procedures, guidelines, and documents that will be followed.
- ❖ **Management:** Identifies the roles, responsibilities, and authorities to execute the plan.
- ❖ **Documentation:** Describes the format, structure, and content of all documents (e.g., test plans, defect reports) to be produced during the software life cycle.
- ❖ **Standards, Practices, and Conventions:** Defines the standards and coding conventions that will be followed.
- ❖ **Reviews and Audits:** Specifies the types and timing of reviews (e.g., design reviews, code reviews) and audits to ensure compliance.
- ❖ **Tools, Techniques, and Methods:** Defines the tools and methods used to support SQA activities (e.g., testing tools, version control tools).

Software Quality Assurance Plan - Contents

- ❖ **Testing:** Outlines the testing strategy and approach, including unit, integration, system, and acceptance testing.
- ❖ **Problem Reporting and Corrective Actions:** Defines procedures for problem tracking and resolution, including the identification of bugs, severity levels, and timelines for corrective actions.
- ❖ **Metrics:** Defines how the quality will be measured and tracked, including defect density, test coverage, and other key performance indicators (KPIs).
- ❖ **Risk Management:** Describes how risks to the software's quality will be identified, assessed, and mitigated.
- ❖ **Training:** Describes training plans for the personnel involved in quality assurance activities.

Software Quality Assurance - Activities in SDLC

- ❖ **SQA** must be integrated in the **SDLC**, not just in the testing phase.
- ❖ **Requirements Phase**
 - **Verification of Requirements:** Ensures that the software requirements are clear, complete, and testable.
 - **Traceability:** Establishes traceability between requirements and design or test cases to ensure all requirements are addressed.
- ❖ **Design Phase**
 - **Design Reviews:** Regular reviews of the design documents to ensure they meet requirements and follow design principles.
 - **Architecture Validation:** Ensures that the software architecture is robust, scalable, and fits the project requirements.

Software Quality Assurance - Activities in SDLC

❖ Implementation Phase

- **Code Reviews:** Systematic examination of source code by peers to identify bugs, vulnerabilities, and violations of coding standards.
- **Version Control and Configuration Management:** Ensures that all versions of the software are controlled and properly documented.

❖ Testing Phase

- **Unit Testing:** Testing of individual components of the software to ensure they function as expected.
- **Integration Testing:** Ensures that the integrated components work together.
- **System Testing:** Evaluates the software as a whole system to verify that it meets its functional and non-functional requirements.
- **Acceptance Testing:** Validates the software against user requirements and ensures it's ready for delivery.
- **Regression Testing:** Ensures that changes made to the software do not introduce new defects.

Software Quality Assurance - Activities in SDLC

❖ Maintenance Phase

- **Post-release SQA:** Monitoring software after release to ensure ongoing quality, bug fixes, and updates follow established standards.
- **Change Management:** Ensures that changes to the software after release are controlled and do not introduce defects.



Software Quality Assurance - Key Concepts

❖ **Software Reviews and Audits**

- **Formal Technical Reviews:** Emphasizes the importance of regular formal technical reviews, such as design reviews, code reviews, and test plan reviews.
- **Audits:** Internal and external audits ensure that the software development processes adhere to organizational and regulatory standards.

❖ **Problem Reporting and Corrective Action**

- **Problems** (such as defects) are identified, documented, and tracked throughout the software lifecycle.
- **Corrective actions** are planned and implemented to fix the problems. The effectiveness of the corrective actions is also assessed.

Software Quality Assurance - Key Concepts

❖ **Metrics and Reporting**

- **Quality Metrics:** Metrics are used to evaluate the quality of both the process and the product.

Common metrics include:

- **Defect density:** Number of defects per size of code (e.g., defects per thousand lines of code).
- **Test coverage:** The percentage of the software that has been tested.
- **Mean Time to Failure (MTTF):** Average time the software operates before encountering a failure.
- **Customer-reported defects:** Issues reported by end users in a live environment.

- **Reporting:** Regular reporting of these metrics helps to monitor the quality of the software over time.

❖ **Risk Management:** SQA involves assessing risks to the project's quality objectives (e.g., tight deadlines, resource shortages, technology risks) and implementing mitigation strategies.

❖ **Continuous Process Improvement:** SQA focuses on continuous process improvement, ensuring that feedback from reviews, audits, and testing is used to refine and improve the development process itself.

Software Quality Assurance - Key Objectives

- ❖ The primary goals of SQA are:
 - **Ensure compliance with requirements:** Verifying that the software meets its functional and non-functional requirements.
 - **Prevention of defects:** Focusing on preventing defects in the software process, not just detecting them after they occur.
 - **Improvement of software processes:** Supporting continuous process improvement, ensuring that feedback is incorporated to enhance software quality in future projects.
 - **Customer satisfaction:** Producing software that meets user needs and provides a high level of satisfaction.

Software Quality Assurance - IEEE Standards

- ❖ In addition to **IEEE 730** which is the main standard that provides a framework and guidelines for SQA through various standards, there are several other standards that relate to different aspects of SQA:
- ❖ **IEEE 1012**: Standard for Software Verification and Validation – focuses on ensuring that the software satisfies its requirements and performs correctly in its intended environment.
- ❖ **IEEE 829**: Standard for Software and System Test Documentation – provides guidelines for documenting the testing process.
- ❖ **IEEE 12207**: Standard for Software Lifecycle Processes – defines the processes involved in software development, including quality assurance activities across the lifecycle.
- ❖ **IEEE 1061**: Standard for Software Quality Metrics Methodology – outlines a methodology for establishing and evaluating software quality metrics.

Software Quality Assurance

❖ Some useful links

- [Software Quality Assurance](#)
- [Quality Assurance, Quality Control, and Testing](#)
- [Software Quality Assurance 101 – The basics](#)
- [What is Quality Assurance\(QA\)?](#)
- [Software Quality Assurance – Software Engineering](#)